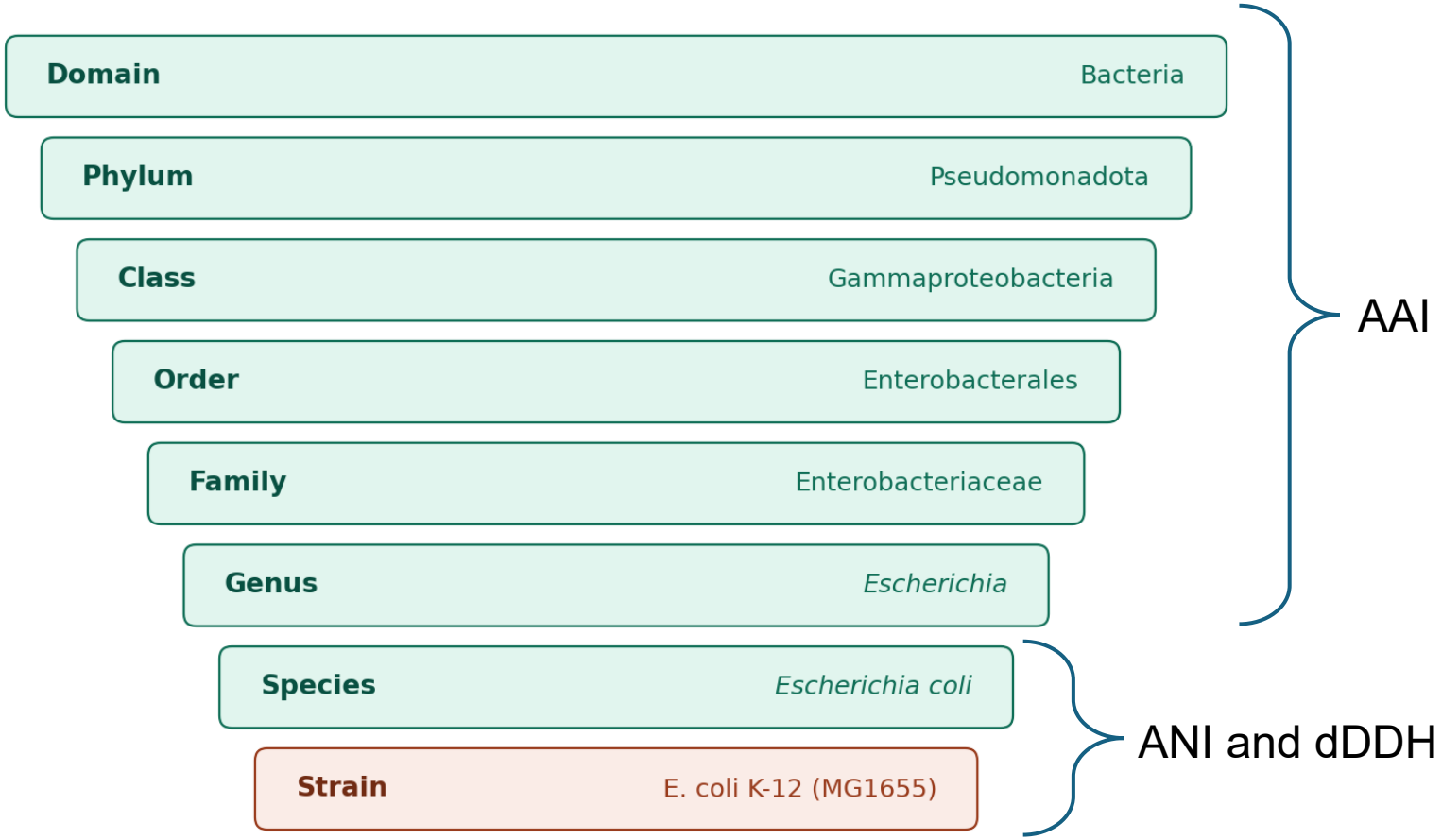
Day 6: Comparative genomics

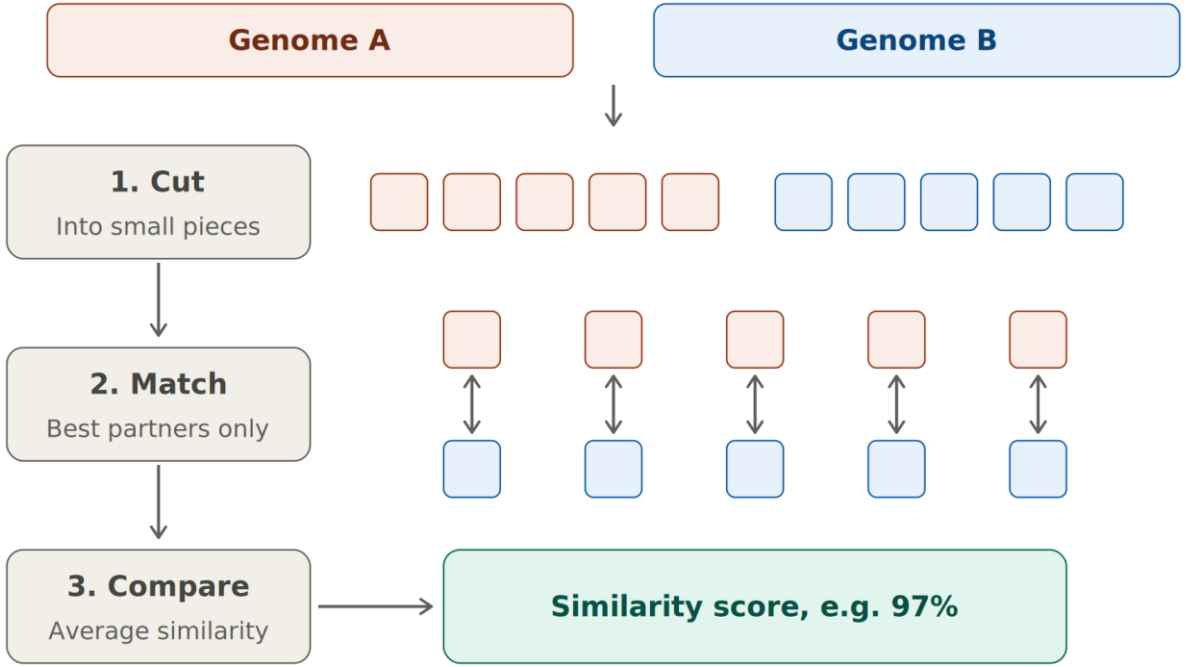
MMB-127

Genome related indexes to classify and compare bacteria



Genome related indexes to classify and compare bacteria

- Average nucleotide identity (ANI)

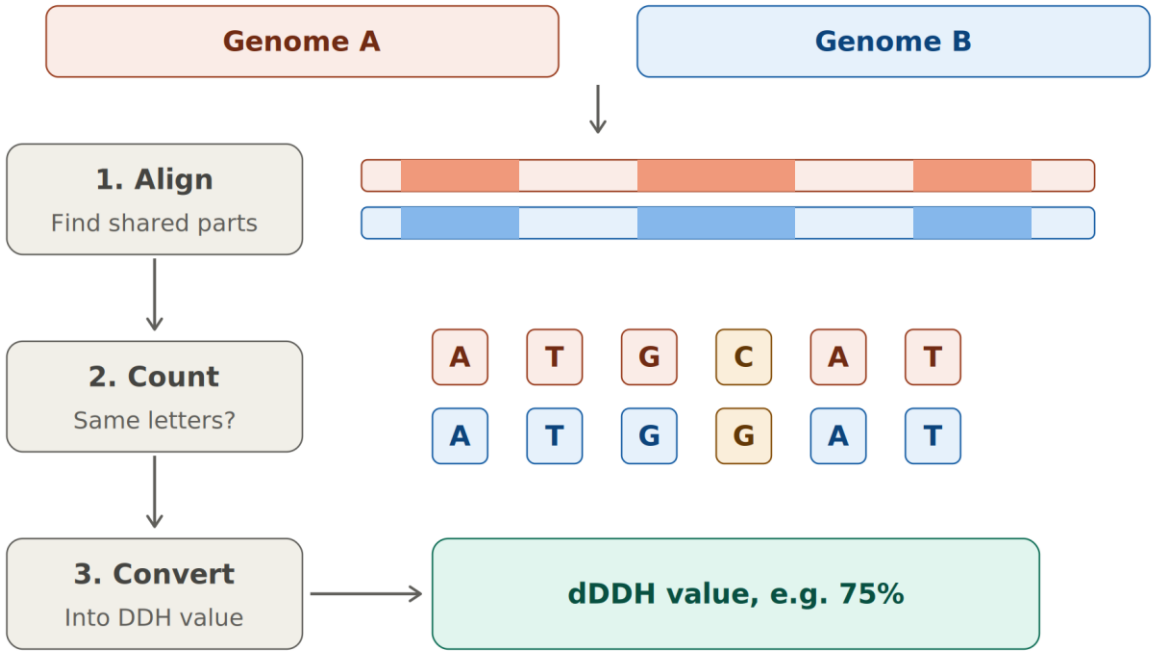


Taxonomic rank	ANI/AAI threshold
(Same) Strain	ANI > 99.99%, gene-content >99% (typical)
Genomovar or sequence type	ANI > 99.5%, gene-content > 90% (typical)
Species	ANI > 95.0%, gene-content > 70% (typical)
Genus	AAI > 65.0%, gene-content > 50% (typical)
Family	AAI > 45.0%, gene-content > 20% (typical)

Konstantinidis, 2023

Genome related indexes to classify and compare bacteria

- Digital DNA-DNA hybridization



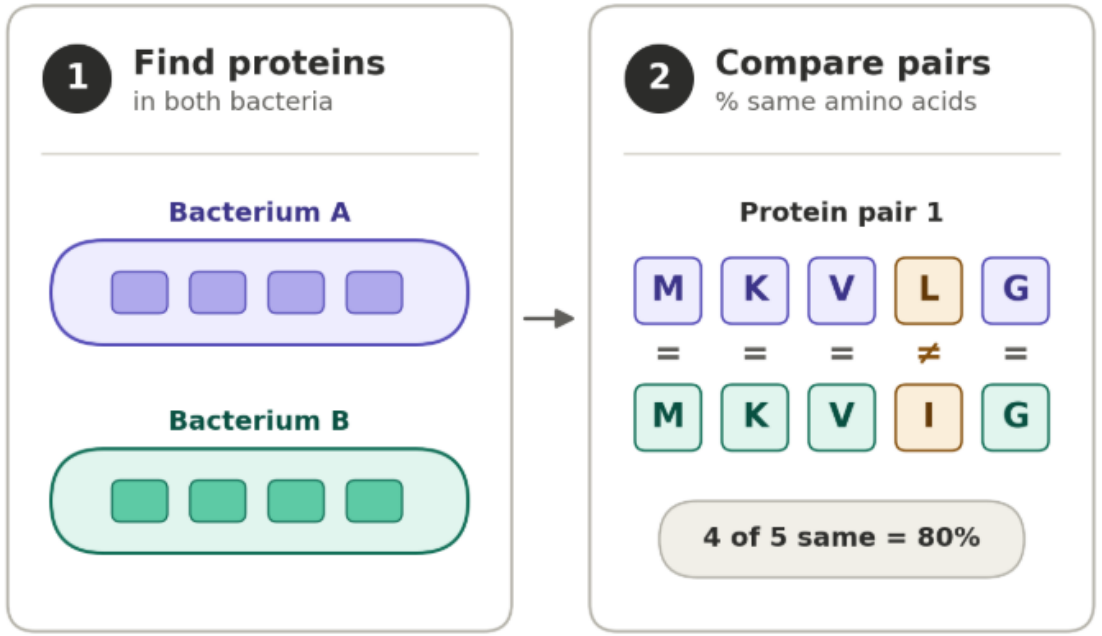
- Formula 1
- Formula 2 = the only one robust to incomplete genomes
- Formula 3 = Complete/closed genomes

>70% = same species

Auch et al., 2010

Genome related indexes to classify and compare bacteria

- Average amino acid identity



60 to 85% for genus delineation

Riesco & Trujillo., 2024

Key words GTDBTK

- sp: It belongs to a distinct species that has no formal name yet.

Example:

```
s__Streptomyces sp000716605
```

- g__Bacillus_A, g__Bacillus_B, g__Bacillus_C. In GTDB, the letter suffix means the old genus was split into several separate genera.

GTDB name	Meaning
`g__Bacillus`	the "true" *Bacillus*, the group containing the type species (*B. subtilis*). It keeps the plain name.
`g__Bacillus_A`	a different genus that used to be called *Bacillus*
`g__Bacillus_B`	another different genus
`g__Bacillus_C`	another one, and so on

Riesco & Trujillo., 2024